

Part 6 Wireshark Analysis

Each exercise captures network traffic and maps it to the OSI model layers while connecting it with distributed system concept

Exercise 6.1 — TCP Three-Way Handshake (Layer 4 — Transport)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	127.0.0.1	127.0.0.1	TCP	56	58269 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
2	0.000107	127.0.0.1	127.0.0.1	TCP	56	8000 → 58269 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
3	0.000155	127.0.0.1	127.0.0.1	TCP	44	58269 → 8000 [ACK] Seq=1 Ack=1 Win=65280 Len=0

Analysis:

1. The SYN packet (packet 1) initiates TCP connections from the client to the server on port 8000.
2. The SYN-ACK (packet 2) acknowledges the client SYN request and responds the request with its own SYN packet.
3. The ACK (packet 3) completes the TCP three-way handshake after the connection is confirmed to be established.

Distributed System Concept: The three-way handshake establishes a reliable connection in distributed systems as the TCP ensures that both the client and the server are prepared to communicate, ensuring ordered and reliable delivery.

Exercise 6.2 — HTTP GET with Full Layer Expansion (Layers 1–7)

```
▼ Frame 4: Packet, 219 bytes on wire (1752 bits), 219 bytes captured (1752 bits) on interface \Device\NPF_{...}, id 0
  Section number: 1
  ▶ Interface id: 0 (\Device\NPF_{...})
    Encapsulation type: NULL/Loopback (15)
    Arrival Time: Jun 23, 2026 04:47:55.794534000 Malay Peninsula Standard Time
    UTC Arrival Time: Jun 22, 2026 20:47:55.794534000 UTC
    Epoch Arrival Time: 1782161275.794534000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 650.000 microseconds]
    [Time delta from previous displayed frame: 650.000 microseconds]
    [Time since reference or first frame: 855.000 microseconds]
    Frame Number: 4
    Frame Length: 219 bytes (1752 bits)
    Capture Length: 219 bytes (1752 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: null:ip:tcp:http]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
  ▼ Null/Loopback
    Family: IP (2)
  ▼ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      Total Length: 215
      Identification: 0x0fde (4062)
    ▶ 010. .... = Flags: 0x2, Don't fragment
      ...0 0000 0000 0000 = Fragment Offset: 0
      Time to Live: 128
      Protocol: TCP (6)
      Header Checksum: 0xec40 [validation disabled]
      [Header checksum status: Unverified]
      Source Address: 127.0.0.1
      Destination Address: 127.0.0.1
      [Stream index: 0]
```

```

Transmission Control Protocol, Src Port: 63391, Dst Port: 8000, Seq: 1, Ack: 1, Len: 175
  Source Port: 63391
  Destination Port: 8000
  [Stream index: 0]
  [Stream Packet Number: 4]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 175]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 3417524163
  [Next Sequence Number: 176 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 472031555
  0101 .... = Header Length: 20 bytes (5)
  Flags: 0x018 (PSH, ACK)
  Window: 255
  [Calculated window size: 65280]
  [Window size scaling factor: 256]
  Checksum: 0x5b7f [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  [Timestamps]
  [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (175 bytes)
Hypertext Transfer Protocol
  GET /api/v1/students/ HTTP/1.1\r\n
  User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; en-MY) WindowsPowerShell/5.1.26100.8655\r\n
  Host: 127.0.0.1:8000\r\n
  Connection: Keep-Alive\r\n
  \r\n
  [Response in frame: 8]
  [Full request URI: http://127.0.0.1:8000/api/v1/students/]

```

OSI Layers: All layers (Layer 1-7)

Layer 1 (Physical): Shows raw transmission details

Layer 2 (Data Link): Used for local loopback traffic

Layer 3 (Network): Network IP

Layer 4 (Transport): TCP

Layer 5 (Session): Session management

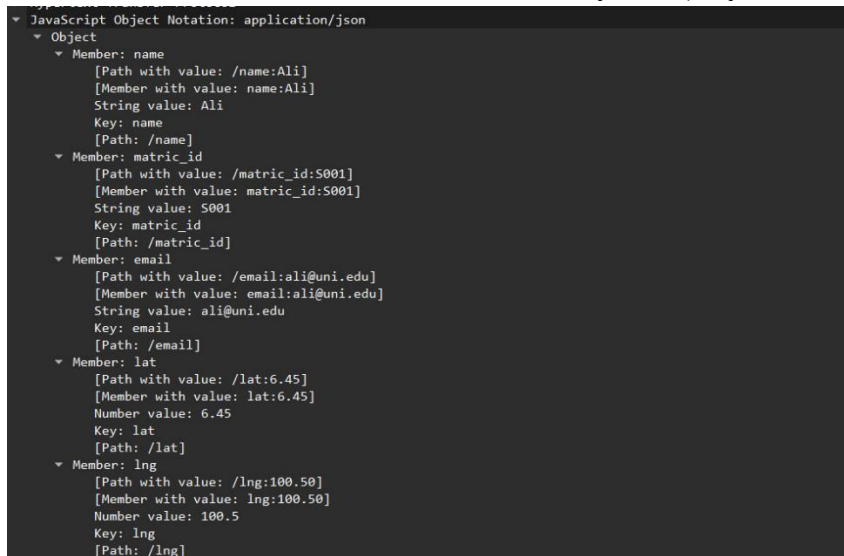
Layer 6 (Presentation): Data representation

Layer 7 (Application): HTTP GET

Analysis: The client sends an HTTP GET request which the server processes before returning the HTTP response. HTTP is processed at the Application Layer (Layer 7).

Distributed System Concept: Each layer adds its own header as data travels down the stack and removes it when arrives. The application layer which is the HTTP GET request is wrapped in TCP and then IP before being wrapped into a data link frame for transmission.

Exercise 6.3 — HTTP POST with JSON Payload (Layer 6 — Presentation)



Analysis: The HTTP POST request is sent to `/api/v1/students/` where Wireshark detects JavaScript Object Notation (JSON) payload. The JSON data was transmitted from the client to the Campus Ride server in plain text.

Distributed System Concept: The JSON represents the presentation layer by converting structured objects into a standardised text format and JSON serves as the data exchange format between the client (browser) and the server (Python FastAPI).

Exercise 6.4 — TCP Four-Way Termination (Layer 4 — Transport)

tcp.flags.fin == 1					
No.	Time	Source	Destination	Protocol	Length Info
10	1.606350	127.0.0.1	127.0.0.1	TCP	44 51913 → 8000 [FIN, ACK] Seq=227 Ack=175 Win=65280 Len=0
12	1.629935	127.0.0.1	127.0.0.1	TCP	44 8000 → 51913 [FIN, ACK] Seq=175 Ack=228 Win=65280 Len=0
23	166.175494	127.0.0.1	127.0.0.1	TCP	44 8000 → 63283 [FIN, ACK] Seq=175 Ack=308 Win=65024 Len=0
36	202.766853	127.0.0.1	127.0.0.1	TCP	44 63372 → 8000 [FIN, ACK] Seq=227 Ack=175 Win=65280 Len=0
38	202.779267	127.0.0.1	127.0.0.1	TCP	44 8000 → 63372 [FIN, ACK] Seq=175 Ack=228 Win=65280 Len=0
48	266.188490	127.0.0.1	127.0.0.1	TCP	44 63283 → 8000 [FIN, ACK] Seq=308 Ack=176 Win=65280 Len=0

Analysis: Packet 10 sends a FIN packet to the server port 8000 and Packet 11 (ACK) acknowledges the client FIN. The packet 12 sends FIN packet to the client and the final ACK indicated that the client acknowledged the server's FIN, completing the TCP four-way termination.

Distributed System Concept: The TCP four-way termination allows both the client and server to finish sending data before the connection is closed. This applies to Campus Ride because the server may send a large response when the client requests a closure to prevent data loss.

Exercise 6.5 — Docker Container-to-Container Traffic (L2/L3 Networking)

```
[
  {
    "Name": "dist_sys_app.default",
    "Id": "98e75095ad1a48ca699561bae6089165097e5079390934ee0f525e88342fa8",
    "Created": "2006-06-22T20:26:03.983358855Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {
      "com.docker.compose.config-hash": "abe8dd938ebteece1afae7466ee61b6ca81e0159df9e104be8f1387b789902",
      "com.docker.compose.network": "default",
      "com.docker.compose.project": "dist_sys_app",
      "com.docker.compose.version": "5.1.1"
    },
    "Containers": {
      "b2b6c6d7a857450d8a0e736dbf7bf5fc459b2768c6971437e89553450c8b6e": {
        "Name": "dist_sys_app.postgres-1",
        "EndpointID": "3ab13a2fb72b2d7bb36f1e4fe5ef3a74d0a1e171d6e9d5a199242b53d0b0d00",
        "MacAddress": "c6:c9:d3:3c:a6:f8",
        "IPv4Address": "172.18.0.2/16",
        "IPv6Address": ""
      },
      "e3ab5d96cc413c3fcfd7376ab2053a2efb13f4e0dfdf13f86d885fc5671fae0": {
        "Name": "dist_sys_app.backend-1",
        "EndpointID": "1389d546fb50f627417fe4df1a439746f8cc167651906d9316783e039ddadacab",
        "MacAddress": "ce:8e:b5:90:41:1f",
        "IPv4Address": "172.18.0.3/16",
        "IPv6Address": ""
      }
    },
    "Status": {
      "IPAM": {
        "Subnets": [
          {
            "172.18.0.0/16": {
              "IPsInUse": 5,
              "DynamicIPsAvailable": 65531
            }
          }
        ]
      }
    }
  }
]
```

Analysis: Both containers communicate internally over the bridge network that was created by Docker. The backend communicates with the PostgreSQL database on port 5432.

Distributed Systems Concept: The Campus Ride Architecture which includes the backend and database is isolated within the Docker private network to ensure that it is not exposed to the public internet.

Exercise 6.6 — TLS/HTTPS Analysis (Railway Production)

tls.handshake					
No.	Time	Source	Destination	Protocol	Length Info
54	3.045733	2405:3800:847:8a22::	2001:4860:482a:7700::	QUIC	1292 Initial, DCID=ceca4aaf529d5, PKN: 3, PING, PADDING, PING, CRYPTO, CRYPTO, PADDING, CRYPTO, CR
92	3.115987	2001:4860:482a:7700::	2405:3800:847:8a22::	QUIC	1292 Initial, SCID=eece4aaf529d5, PKN: 5, CRYPTO, PADDING
186	3.467026	2405:3800:847:8a22::	2404:6800:4001:813::	QUIC	1292 Initial, DCID=5baef4c731d7214, PKN: 3, PING, PADDING, PING, PING, PADDING, CRYPTO, CRYPTO, PING,
197	3.534594	2404:6800:4001:813::	2405:3800:847:8a22::	QUIC	1292 Initial, SCID=fbaef4c731d7214, PKN: 5, CRYPTO, PADDING
213	3.566175	2405:3800:847:8a22::	2404:6800:4001:813::	QUIC	1292 Initial, DCID=3e8b6a5193c177ed, PKN: 3, PING, CRYPTO, PADDING, CRYPTO, PADDING, CRYPTO, PADDING,
220	3.628810	2404:6800:4001:813::	2405:3800:847:8a22::	QUIC	1292 Initial, SCID=f8b6a5193c177ed, PKN: 5, CRYPTO, PADDING
273	4.534524	2405:3800:847:8a22::	2404:6800:4001:80d::	QUIC	1292 Initial, DCID=d711be97e60ce94a, PKN: 1, PADDING, PING, CRYPTO, CRYPTO, PING, PING, PING, CRYPTO,
278	4.534878	2405:3800:847:8a22::	2404:6800:4001:80d::	QUIC	1292 Initial, DCID=d711be97e60ce94a, PKN: 2, PADDING, CRYPTO, PING, PING, PADDING, CRYPTO, PING, PING,
282	4.598071	2404:6800:4001:80d::	2405:3800:847:8a22::	QUIC	1292 Initial, SCID=f711be97e60ce94a, PKN: 3, CRYPTO, PADDING
386	5.556509	2405:3800:847:8a22::	2a04:4e42:76::347	TLSv1.3	590 Client Hello (SNI=ohhttp-relay-safebrowsing-chrome.google.fastly-edge.com)
388	5.592289	2a04:4e42:76::347	2405:3800:847:8a22::	TLSv1.3	1386 Server Hello, Change Cipher Spec, Application Data
410	5.779638	10.30.234.79	69.46.46.47	TLSv1.3	435 Client Hello (SNI=frontend-production-974f.up.railway.app)
414	5.801786	10.30.234.79	69.46.46.47	TLSv1.3	499 Client Hello (SNI=frontend-production-974f.up.railway.app)
417	6.047487	69.46.46.47	10.30.234.79	TLSv1.3	1454 Server Hello, Change Cipher Spec
422	6.048503	10.30.234.79	69.46.46.47	TLSv1.3	467 Client Hello (SNI=frontend-production-974f.up.railway.app)
428	6.081405	69.46.46.47	10.30.234.79	TLSv1.3	1454 Server Hello, Change Cipher Spec
434	6.330824	69.46.46.47	10.30.234.79	TLSv1.3	1454 Server Hello, Change Cipher Spec

Analysis: Client Hello (TLSv1.3) was sent to the Railway server along with the SNI field which is frontend-production-974f.up.railway.app. The server responds with the selected

cipher suite. The actual HTTP content for the application data was encrypted and rendered unreadable using TLS.

Distributed System Concept: TLS is a process at the presentation layer that encrypts and decrypts data before it reaches the application layer. In this Campus Ride, TLS enables a shared secret between client and server without transmitting the secret.